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Date: 15-3-206

Purple Team Operations: Strategic Reconnaissance & Defensive IDS Integration

1.0 Project Introduction

1.1 Project Summary

The main purpose of this project was to create a secure and isolated laboratory environment for practicing real-world cybersecurity techniques. During this phase, the focus was on Passive Reconnaissance, Network Traffic Analysis, and Active Vulnerability Scanning.

A dual virtual machine setup consisting of Kali Linux and Windows 10 was configured to simulate an attacker–target environment. Using this setup, the network structure was analyzed and the attack surface of the target system was successfully identified.

Key outcomes of this phase include:

- **Protocol Analysis:**

  The TCP Three-Way Handshake was captured and examined to understand the connection establishment process.

- Vulnerability Identification: An open SMB service on port 445 was discovered on the target system with IP address 192.168.88.1

- Network Monitoring Status: It was verified that the network monitoring engine is correctly capturing and processing packets through the eth0 network interface.

2.0 Lab Environment Setup

2.1 Virtual Network Architecture

The cybersecurity lab environment was created using VMware Workstation with a dedicated internal network segment.

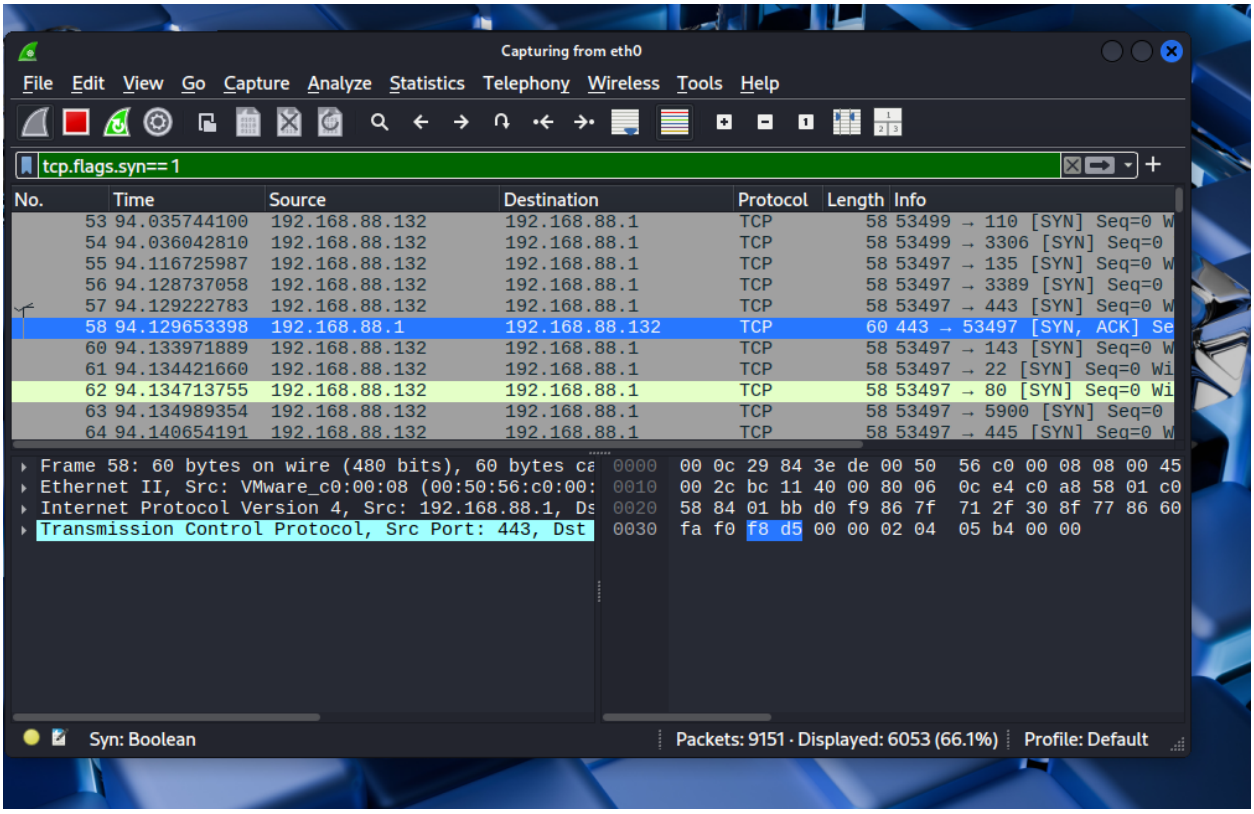
This configuration ensures that all testing activities, including scanning and simulated attack traffic, remain isolated within the virtual network and do not impact the host computer or external networks.

2.2 System and Network Configuration

The lab environment consists of two virtual machines configured as follows:

- Attacker System: Kali Linux – 192.168.88.1

To confirm proper connectivity between both machines, a network test was performed using the Test-NetConnection utility. This step ensured that the network communication and system connectivity were functioning correctly before beginning further analysis and scanning activities.



3.0 Passive Information Gathering

3.1 Open-Source Intelligence (OSINT) Investigation

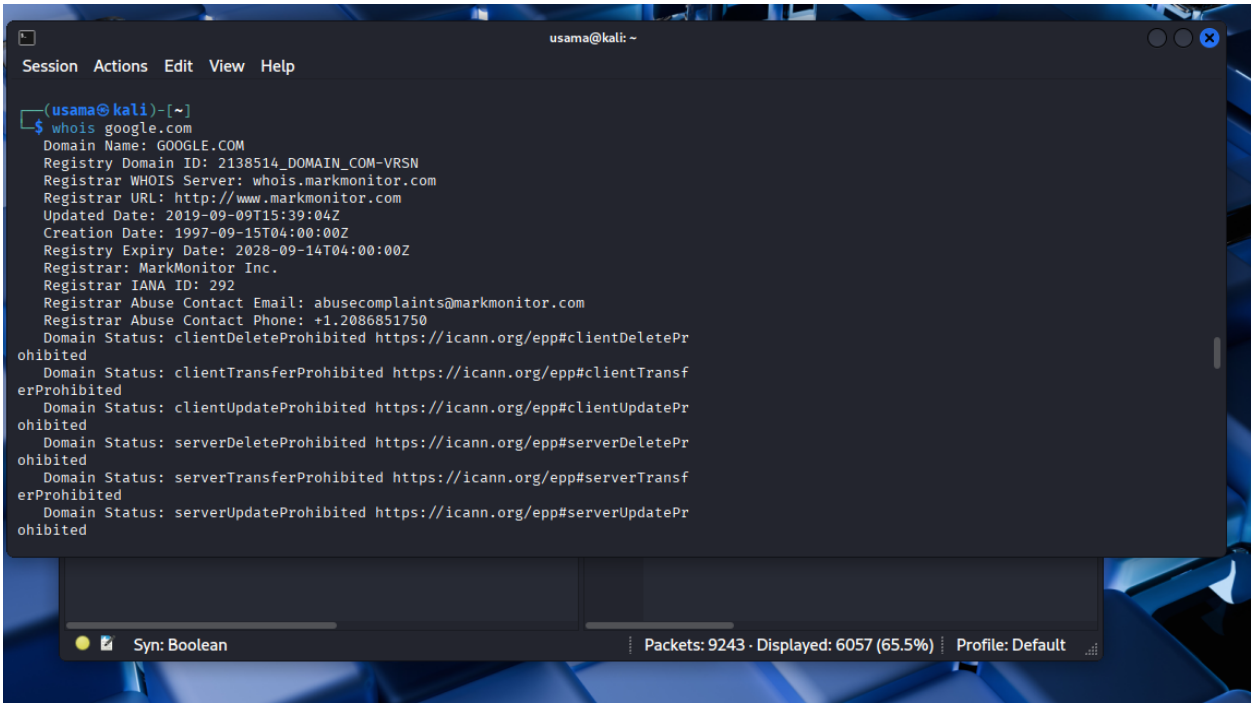
Passive reconnaissance is an essential initial stage in cybersecurity assessments. In this phase, the **whois utility** was used to collect publicly available information about the target domain without directly communicating with its servers.

This method allows analysts to gather details related to **domain ownership, registration information, and infrastructure components** while remaining undetected by the target system.

3.2 Domain Registration Examination

The collected registry information was analyzed to understand the structure and background of the target domain.

- **Target Domain:** https://www.google.com/search?q=google.com
- **Infrastructure Details:** The analysis revealed the presence of **four authoritative Name Servers (ns1, ns2, ns3, and ns4)** responsible for managing domain resolution.
- **Security & Forensic Insight:** Examining domain **registration and expiration dates** can help security analysts evaluate the **stability, age, and potential security posture** of the domain's infrastructure.



4.0 Network Traffic Monitoring

4.1 Real-Time Packet Inspection

For analyzing network communication between the two virtual machines, **Wireshark** was used to capture live packets. This tool allows detailed inspection of network traffic across multiple protocol layers including **Ethernet, IP, and TCP**.

Through this packet capture process, the interaction between the attacker system and the target machine was monitored in real time.

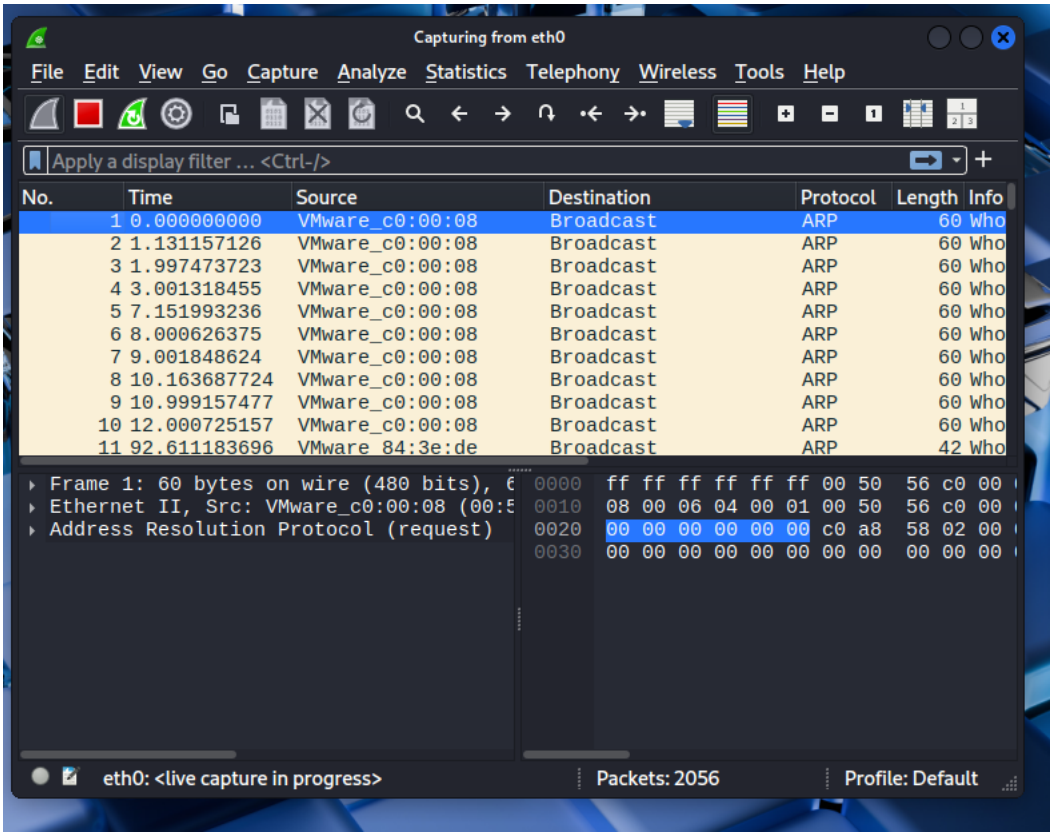
4.2 TCP Connection Analysis

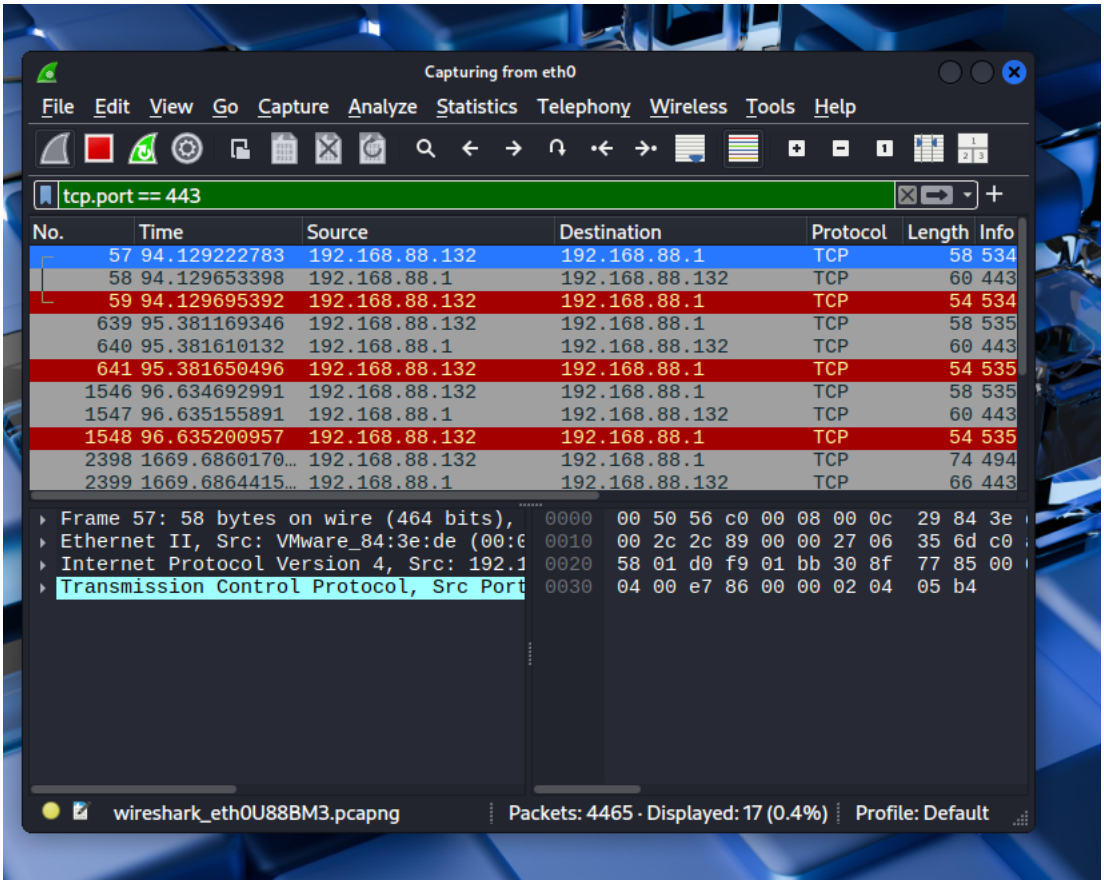
To analyze the connection establishment process, a display filter was applied to focus on the packets responsible for starting a TCP session.

- **Filter Applied:** tcp.flags.syn == 1
  This filter helps isolate packets that initiate new TCP connections.

Connection Establishment Process

1. **SYN Packet:**
   The source machine (**192.168.220.128**) sent a synchronization request to **port 445** on the target system.
2. **SYN-ACK Packet:**
   The destination host (**192.168.220.131**) responded with a synchronization acknowledgment, indicating that the requested port is active and ready to accept connections.
3. **ACK Packet:**
   The source system sent the final acknowledgment packet, completing the **TCP three-step handshake** and successfully establishing the connection.


Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 VMware_c0:00:08 Broadcast ARP 60 Who
2 1.131157126 VMware_c0:00:08 Broadcast ARP 60 Who
3 1.997473723 VMware_c0:00:08 Broadcast ARP 60 Who
4 3.001318455 VMware_c0:00:08 Broadcast ARP 60 Who
5 7.151993236 VMware_c0:00:08 Broadcast ARP 60 Who
6 8.000626375 VMware_c0:00:08 Broadcast ARP 60 Who
7 9.001848624 VMware_c0:00:08 Broadcast ARP 60 Who
8 10.163687724 VMware_c0:00:08 Broadcast ARP 60 Who
9 10.999157477 VMware_c0:00:08 Broadcast ARP 60 Who
10 12.000725157 VMware_c0:00:08 Broadcast ARP 60 Who
11 92.611183696 VMware_84:3e:de Broadcast ARP 42 Who
Frame 1: 60 bytes on wire (480 bits), 6
Ethernet II, Src: VMware_c0:00:08 (00:5
Address Resolution Protocol (request)
0000 ff ff ff ff ff ff 00 50 56 c0 00
0010 08 00 06 04 00 01 00 50 56 c0 00
0020 00 00 00 00 00 00 c0 a8 58 02 00
0030 00 00 00 00 00 00 00 00 00 00 00
eth0: <live capture in progress>
Packets: 2056
Profile: Default



5.0 Active Network Scanning

5.1 Port and Service Identification Approach

To identify accessible services on the target system, an **active network scan** was conducted using **Nmap**.
This scanning technique helps security analysts detect **open ports and determine the versions of services** running on a target host.

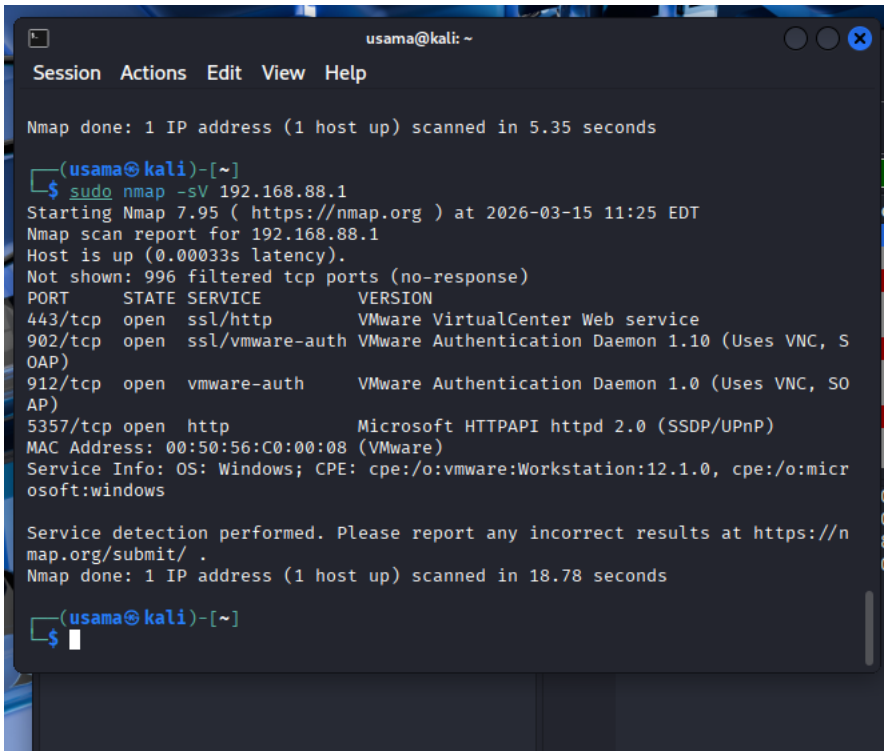
5.2 Scan Results and Observations

The scanning process produced the following technical findings:

- **Open Port Detection:**
  The scan revealed that **port 445/TCP** is open on the target machine.
- **Service Identification:**
  The running service was recognized as **microsoft-ds**, which is typically

associated with the **SMB (Server Message Block)** protocol used for file and printer sharing in Windows systems.

- **Verification Method:**
  A **TCP Connect scan (-sT)** was performed to confirm the port status. This scan establishes a complete TCP connection with the target host, ensuring accurate verification of the service availability.



```
usama@kali: ~
Session  Actions  Edit  View  Help

Nmap done: 1 IP address (1 host up) scanned in 5.35 seconds

┌──(usama㉿kali)-[~]
└─$ sudo nmap -sV 192.168.88.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 11:25 EDT
Nmap scan report for 192.168.88.1
Host is up (0.00033s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE SERVICE         VERSION
443/tcp  open  ssl/http        VMware VirtualCenter Web service
902/tcp  open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, S
OAP)
912/tcp  open  vmware-auth     VMware Authentication Daemon 1.0 (Uses VNC, SO
AP)
5357/tcp open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 00:50:56:C0:00:08 (VMware)
Service Info: OS: Windows; CPE: cpe:/o:vmware:Workstation:12.1.0, cpe:/o:micr
osoft:windows

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.78 seconds

┌──(usama㉿kali)-[~]
└─$
```

```
└─$ sudo nmap -sT -Pn -p 445 192.168.220.131
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 08:50 EDT
Nmap scan report for 192.168.220.131
Host is up (0.00079s latency).
```

6.0 Security Monitoring and Detection

6.1 Intrusion Detection System Setup

In Purple Team exercises, it is important that offensive testing activities can be identified by defensive security mechanisms. For this purpose, the **Intrusion Detection System (IDS)** was initialized to monitor network activity.

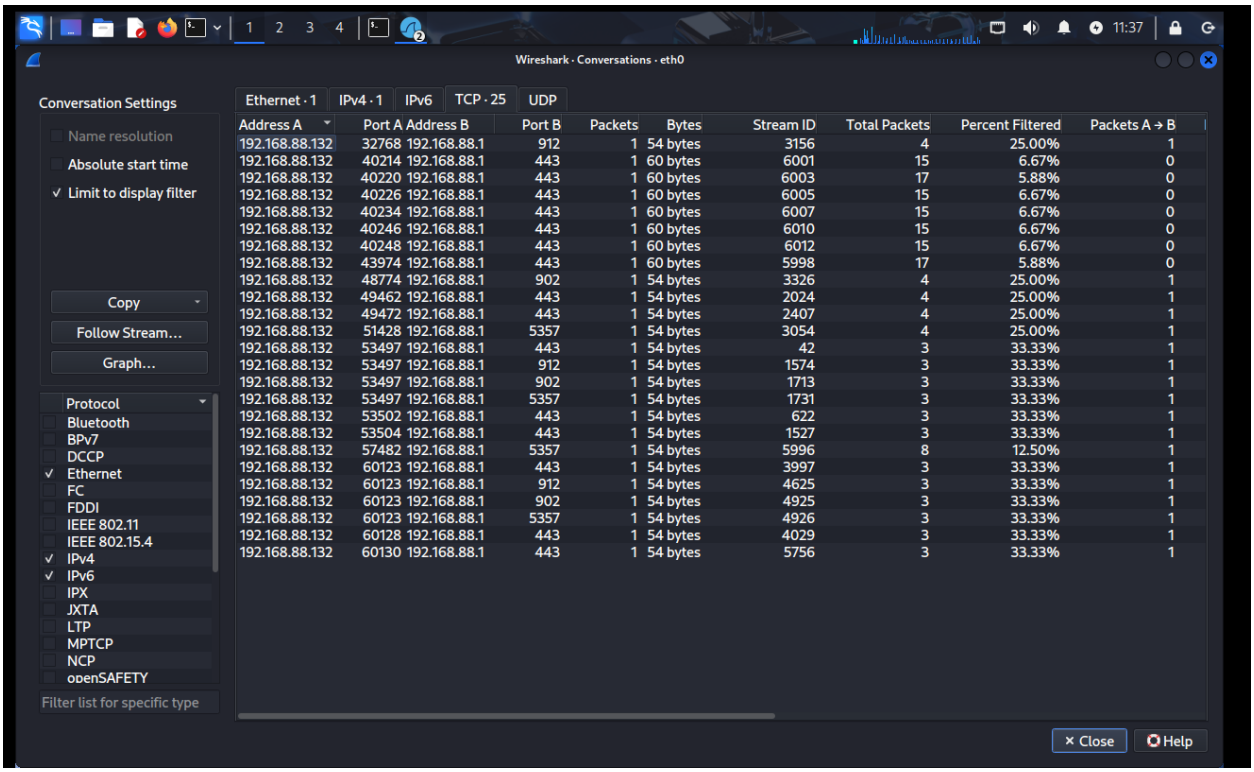
The system was configured to observe the **eth0 network interface**, enabling real-time inspection of packets flowing across the network during the scanning process.

6.2 Log Monitoring and Verification

The monitoring logs were analyzed to ensure that the detection system was functioning correctly.

- **Operational Mode:**
  The monitoring engine was configured in **Passive Mode**, which allows it to listen to all network traffic within the segment without actively modifying or blocking the data flow.

- **System Status:**
  Log entries indicated the message **"Commencing packet processing,"** confirming that the monitoring system had successfully started capturing and analyzing network packets generated during the scanning activities.



Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B
192.168.88.132	32768	192.168.88.1	912	1	54 bytes	3156	4	25.00%	1
192.168.88.132	40214	192.168.88.1	443	1	60 bytes	6001	15	6.67%	0
192.168.88.132	40220	192.168.88.1	443	1	60 bytes	6003	17	5.88%	0
192.168.88.132	40226	192.168.88.1	443	1	60 bytes	6005	15	6.67%	0
192.168.88.132	40234	192.168.88.1	443	1	60 bytes	6007	15	6.67%	0
192.168.88.132	40246	192.168.88.1	443	1	60 bytes	6010	15	6.67%	0
192.168.88.132	40248	192.168.88.1	443	1	60 bytes	6012	15	6.67%	0
192.168.88.132	43974	192.168.88.1	443	1	60 bytes	5998	17	5.88%	0
192.168.88.132	48774	192.168.88.1	902	1	54 bytes	3326	4	25.00%	1
192.168.88.132	49462	192.168.88.1	443	1	54 bytes	2024	4	25.00%	1
192.168.88.132	49472	192.168.88.1	443	1	54 bytes	2407	4	25.00%	1
192.168.88.132	51428	192.168.88.1	5357	1	54 bytes	3054	4	25.00%	1
192.168.88.132	53497	192.168.88.1	443	1	54 bytes	42	3	33.33%	1
192.168.88.132	53497	192.168.88.1	912	1	54 bytes	1574	3	33.33%	1
192.168.88.132	53497	192.168.88.1	902	1	54 bytes	1713	3	33.33%	1
192.168.88.132	53497	192.168.88.1	5357	1	54 bytes	1731	3	33.33%	1
192.168.88.132	53502	192.168.88.1	443	1	54 bytes	622	3	33.33%	1
192.168.88.132	53504	192.168.88.1	443	1	54 bytes	1527	3	33.33%	1
192.168.88.132	57482	192.168.88.1	5357	1	54 bytes	5996	8	12.50%	1
192.168.88.132	60123	192.168.88.1	443	1	54 bytes	3997	3	33.33%	1
192.168.88.132	60123	192.168.88.1	912	1	54 bytes	4625	3	33.33%	1
192.168.88.132	60123	192.168.88.1	902	1	54 bytes	4925	3	33.33%	1
192.168.88.132	60123	192.168.88.1	5357	1	54 bytes	4926	3	33.33%	1
192.168.88.132	60128	192.168.88.1	443	1	54 bytes	4029	3	33.33%	1
192.168.88.132	60130	192.168.88.1	443	1	54 bytes	5756	3	33.33%	1

7.0 Security Assessment and Final Evaluation

7.1 Discovered Network Exposure

During the scanning phase, **Port 445** was found to be open on the Windows target system. This discovery is considered important because this port is commonly associated with the **SMB (Server Message Block)** protocol used for network file sharing.

Historically, this port has been targeted by various cyber threats such as **ransomware attacks and lateral movement techniques**, making it a potential entry point if not properly secured.

7.2 Recommended Security Measures

To reduce potential security risks, the following actions are recommended:

- **Close Unnecessary Ports:**
  If SMB services are not required, **port 445** should be disabled or blocked through the system's firewall settings.
- **Apply Security Updates:**
  The operating system should be regularly updated to protect against known vulnerabilities, including exploits such as **MS17-010**.
- **Continuous Traffic Monitoring:**
  The deployed packet monitoring engine should be used to detect abnormal network activities, such as repeated **SYN scanning attempts**, which may indicate reconnaissance or intrusion attempts.

7.3 Conclussion

All the planned objectives for **Week 01 and Week 02** were successfully achieved. The lab environment was properly configured, network connectivity was verified, and the attack surface of the target system was identified.
Additionally, the defensive monitoring mechanism was validated, confirming that network traffic generated during the scanning activities can be effectively observed and analyzed. The lab environment is now prepared for the **next stage of security testing and exploitation analysis**.